

Informatics 141

Computer Science 121

Information Retrieval

Lecture 5

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These course materials borrow, with permission, from those of Prof. Cristina Videira Lopes, Prof. Alberto Krone-Martins, Addison Wesley 2008, Chris Manning, Pandu Nayak, Hinrich Schütze, Heike Adel, Sascha Rothe, Jerome H. Friedman, Robert Tibshirani, and Trevor Hastie. Powerpoint theme by Prof. André van der Hoek.

A bit further on Web users:

How users evaluate search engines

Information Retrieval

How do users evaluate search engines?

- Major standard: **Quality of pages**

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How do users evaluate search engines?

- Major standard: **Quality of pages**
 - Classic IR relevance: $R(Q,D)$
 - How you really define relevance is among your engine secrets $R(Q,D,...)$.
 - Also important:
 - Trust
 - Duplicate elimination (*with “show similar” option)
 - Readability
 - Fast access
 - No pop-ups

How do users evaluate search engines?

- **First** : what is **precision** and **recall**?

How do users evaluate search engines?

- **Precision:** *How relevant are the first few hits?*
- **Recall:** *How many relevant hits are presented?*

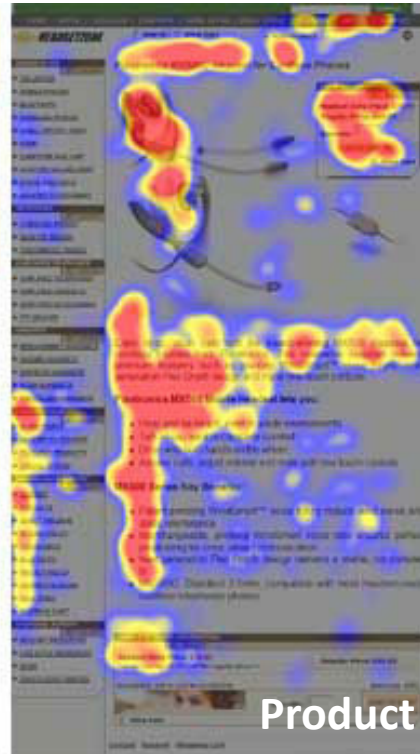
Which one is more important for a web search engine?

How do users evaluate search engines?

- Precision: *How relevant are the first few hits?*
- Recall: *How many relevant hits are presented?*
- **For web search: Precision is more important than recall**

How do users evaluate search engines?

- For web search: Precision is more important than recall



Eyetracking by Nielsen Norman Group nngroup.com NN/g

F-Shape reading patterns

How do users evaluate search engines?

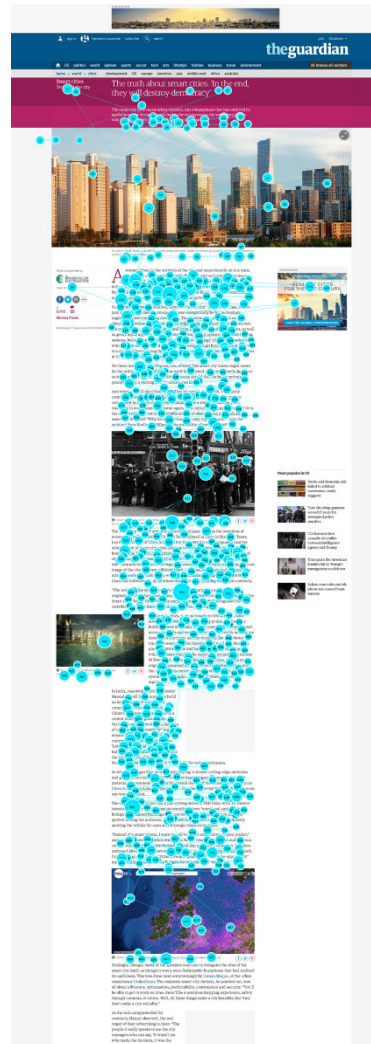
- Precision: *How relevant are the first few hits?*
- Recall: *How many relevant hits are presented?*
- For the web: Precision is more important than recall
- **When is recall important?**

How do users evaluate search engines?

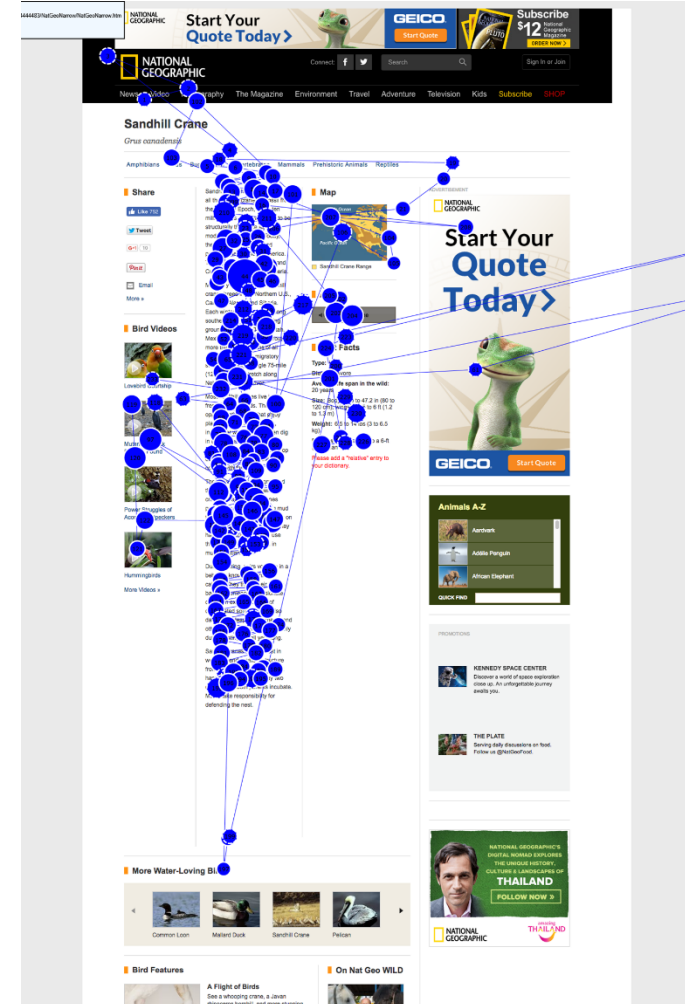
- **When is recall important?** When completeness is required

Commitment reading pattern :
sources trusted *a priori*,
motivated users.

Ex. Search engines
for Law, Sciences.



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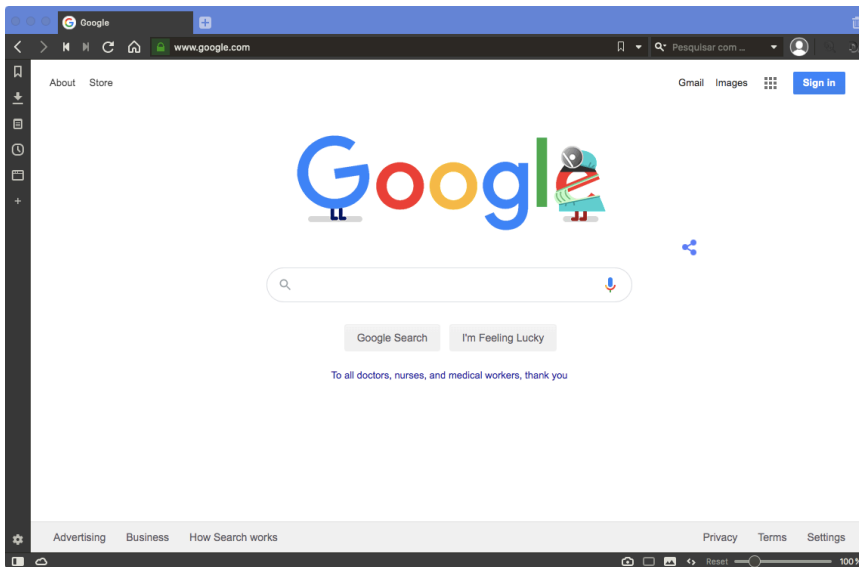
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How do users evaluate search engines?

- Good UI: simple, no clutter

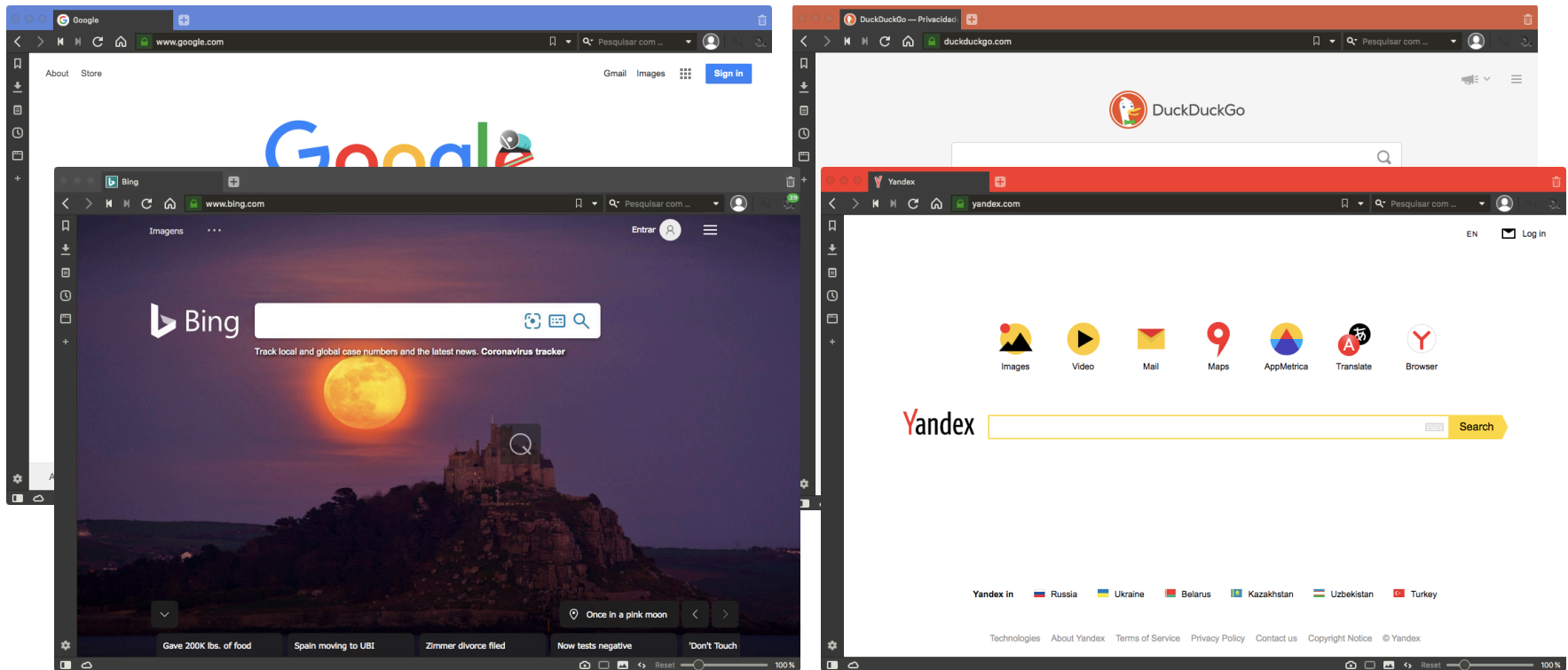
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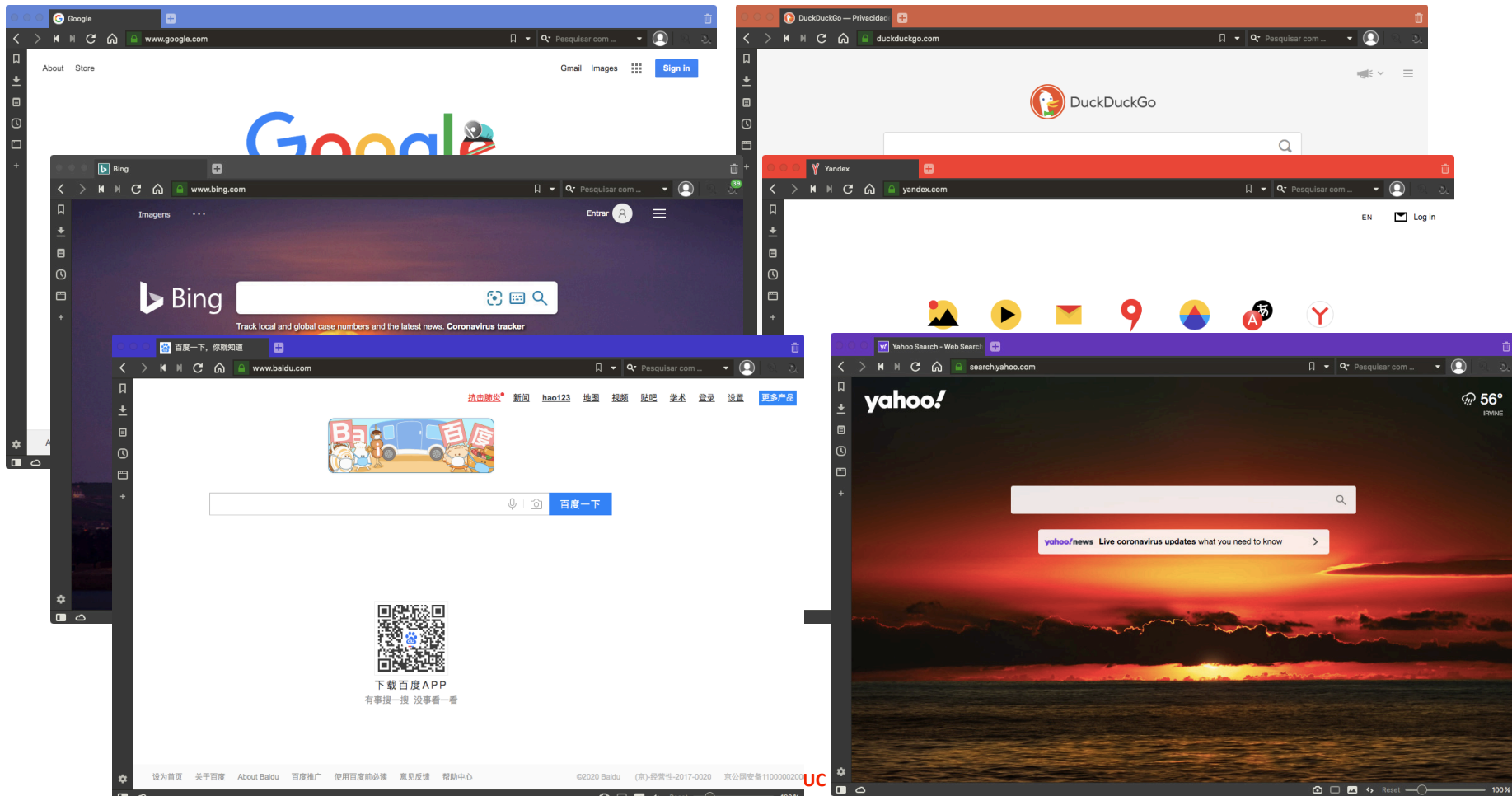
How do users evaluate search engines?

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How do users evaluate search engines?

- Good UI: simple, no clutter



How do users evaluate search engines?

- Good UI: simple, no clutter
- Pre and post processing tools
 - Spell check / auto correct
 - Suggested alternative searches
 - Links to resources (maps, images, etc)
 - Able to deal with syntactic cues
 - ex: URL typed in search box, Math equation, etc.

Web Search business model : Advertising

Information Retrieval



- Early synergy between search and ads
- First: keyword-based engines
 - Altavista (1995-1997), Excite, InfoSeek, Inktomi
- Paid placement ranking
 - Ads based on auction for keyword
 - Goto.com => Overture.com => Yahoo! => Google

- Link-based ranking pioneered by Google (1998+)
 - Links added the idea of authoritativeness to relevance
 - Blew away keyword-only engines
 - Great user experience, **looking for business model**
 - **Meanwhile, Goto/Overture's annual revenues near \$1B**



- Google added paid placement ads on site
 - Differentiated from search results
- Yahoo! build a similar engine
 - Bought Inktomi for search tech (a few hundred million USD!)
 - Bought Overture for paid placement tech (1.6 billion USD in 2003!)
 - Sponsored links placed alongside objective results.
- Google licensed paid placement tech from Yahoo!
 - Google's ad placement engine for many years (perhaps still today, nobody knows with certainty)



- Google and most other search engines maintain that ads do not affect vendors' ranking in search
 - **But again: nobody knows for sure...**
- Ad placement:
 - Fully automated
 - Balance auction price and relevance
 - Targeted advertising : focus on audience

How companies pay for Ads on Search?

- **Cost Per Mil (CPM)**
 - Cost for showing the ad with 10^3 -page shows
 - Important for branding campaigns
- **Cost Per Click (CPC)**
 - Cost for users clicking on the ad
 - Important for sales campaigns
- There are also other concepts
 - e.g. CPA: Cost per action/acquisition; if that click triggered some action; e.g. DuckDuckGo + Amazon.

Warning: Click Fraud

- \$7.2 billion USD were lost between 2016-2018 (Guardian)
- Google calls this “invalid clicks”
- Search engines have sophisticated tools to detect fraud

Information Retrieval

Search Engine Optimization

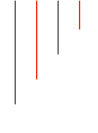
Paid Ad Placement

- It costs money
- So...

Search Engine Optimization (SEO)



- **Tuning of a web page/site/app to rank highly** in search results for certain queries
- Alternative to paying for ad placement
 - Bonus: **visibility through rank = greater trust**
- **It's marketing**: getting your content to your audience



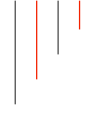
- Motives
 - Commercial
 - Political
 - Religious
 - Lobbying



- Motives
 - Commercial
 - Political
 - Religious
 - Lobbying
- Who does this?
 - Internally: webmasters, writers
 - Commercially: companies, consultants
 - Hosting services
 - Plugins to popular CMSs

- Essentially a **marketing – technology interface** area
- How to do it:
 - <https://support.google.com/webmasters/answer/40349?hl=en>
 - <http://freetools.webmasterworld.com/category/seo-tools>
 - Many books. E.g. :

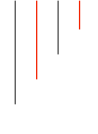




- Ethical and unethical ways of doing it
- Legitimate approach:
 - Indexed age of the pages (older is better)
 - Good incoming links
 - Good content, well written, well organized, up to date
 - Good use of web standards/practices
 - Fast servers (quick response)



- **Unethical approaches (aka spam):**
 - fake pages
 - fake sites that point to your site
 - fake comments/engagement (bots!)
 - in short: “alternative facts” aka lies
- Sometimes the line between legitimate and illegitimate practices is hard to find. There’s a large grey area



- Search results (also) depend on which data center receives the query:
 - google.com vs. google.fr vs google.pt will show different results for a single query: e.g. “Paris”
- Different SEO strategies can be considered per country